LING285-L8

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Section 1: A Spring Alone Versus a Mass-Spring System

- **A spring stops at a place, but a mass stops after a process.** This single idea separates two very different kinds of motion. A spring with no mass attached returns to one location and stays there. A spring with a mass attached keeps moving past that location, back and forth, before it finally comes to rest.
    - Everything here is measured with one number, the **displacement** x. Displacement is how far the end of the spring is from its **natural position**, the place where the spring rests when nothing is pulling or pushing on it. We call the natural position $x = 0$, a stretch to the right $x = +1$, and a squeeze to the left $x = -1$.
- **The spring alone, pulled out.** Picture a spring fixed to a wall at one end, lying on a table, with nothing attached to its free end. If we pull the free end out to $x = +1$ and let go, the spring pulls itself back and stops exactly at $x = 0$. It does not overshoot.
    - When we plot x against time for this release, the curve starts at $+1$ and bends downward, dropping quickly at first and then more and more slowly. The curve gets closer and closer to 0 but never crosses below it. The label for this motion is **"x approaches 0 without crossing it."**
    - The reason the motion slows down is that the pull of the spring depends on how far it is stretched. A large stretch produces a large pull back, and a small stretch produces a small pull. As the end gets close to $x = 0$, the pull becomes almost zero, so the motion becomes almost zero too.
- **The spring alone, pushed in.** The same thing happens in the opposite direction. If we squeeze the spring to $x = -1$ and release it, the spring pushes itself back out and settles at $x = 0$.
    - Plotted against time, this curve starts at -1 and rises toward 0, again quickly at first and then more slowly. It flattens out right at 0 and never goes above it. The label is **"x rises toward 0 without crossing it."**
    - Whether we start from $+1$ or -1, a massless spring always ends at the same place and never passes through it. That is what it means to say a spring **stops at a place**.
- **Adding a mass changes the behavior completely.** Now a mass, drawn as a small cart on wheels, is attached to the free end of the spring. We pull the mass to $x = +1$ and release it. The mass does not stop at $x = 0$. It shoots right through the natural position and keeps going to the other side.
    - The mass keeps going because a moving mass carries its motion with it. When the mass reaches $x = 0$, the spring is no longer pulling on it, but the mass is already moving fast, so it continues past 0 and compresses the spring on the other side.
    - Once the spring is compressed, it pushes the mass back. The mass slows down, stops briefly near $x = -1$, and then gets pushed back through $x = 0$ toward the positive side again. The label for this is **"the mass carries motion through every crossing."**
    - Plotted against time, x now makes a wave. It starts at $+1$, falls through 0 to nearly -1, rises back through 0 to nearly $+1$, and repeats. This repeating back-and-forth motion is called **oscillation**.
- **Lightly damped oscillation.** Each swing of the mass is slightly smaller than the one before it, so the peaks of the wave shrink a little over time. This is called a **lightly damped oscillation**. **Damping** is anything that slowly drains away the motion, such as friction. With light damping, the mass swings many times before it finally settles at $x = 0$.
    - This is why a mass **stops after a process**. It does eventually come to rest at the natural position, but only after a long sequence of crossings back and forth, each one a little smaller than the last.
    - A mass-spring system therefore has a **frequency**, which is the number of back-and-forth cycles it completes per second. A massless spring has no frequency at

all, because it never swings.

Section 2: Perturbation Theory

- **Perturbation theory** asks what happens to the behavior of a system when we change one of its properties. To **perturb** a system means to nudge one of its settings. For a mass-spring system, the two settings that matter are the **mass** m and the **stiffness** k of the spring, and the behavior we care about is the frequency f of the oscillation.
- **Increasing the mass decreases the frequency.** If we attach a heavier mass to the same spring, it oscillates more slowly, so the frequency goes down.
    - A heavier mass is harder to speed up and harder to slow down. The spring pulls on it with the same strength as before, but that pull changes the heavier mass's speed less. Each swing takes longer, so fewer swings fit into one second.
    - In short form: when m goes up, f goes down.
- **Increasing the stiffness increases the frequency.** If we keep the same mass but use a stiffer spring, the oscillation speeds up, so the frequency goes up.
    - **Stiffness** k is how strongly a spring resists being stretched or squeezed. A stiffer spring pulls back harder for the same amount of stretch. That stronger pull speeds the mass up faster, so each swing finishes sooner and more swings fit into one second.
    - In short form: when k goes up, f goes up.
- These two rules together are the core of perturbation theory for oscillators. **More mass means lower frequency, and more stiffness means higher frequency.** We do not need to calculate an exact frequency to use these rules. We only need to know which direction the frequency moves when mass or stiffness changes.

Section 3: Speech as Changing Mass and Stiffness of Air

- **Speaking changes the effective mass and stiffness of air in the vocal tract.** The **vocal tract** is the air-filled tube running from the vocal folds in the throat up through the mouth to the lips. When we speak, we move the tongue, jaw, and lips, and these movements squeeze some portions of the air in that tube and open up others.
    - Squeezing a portion of air in one place changes how much that air acts like a mass and how much it acts like a spring. The **effective mass** and **effective stiffness** of each portion of air are different for every vowel and every consonant, because each sound uses a different shape of the tube.
    - The air in the vocal tract is set into vibration by the vibration of the vocal folds. Because the tube has different mass and stiffness in different places, the air vibrates at **many frequencies at once**. The particular set of frequencies depends on **where** the air is squeezed and **how much** it is squeezed, so each vowel and consonant has its own set.
- **MRI images of five vowels make these shapes visible.** Each image is a side view of the head sliced down the middle, taken while a speaker says one word. The bright, rounded mass in the middle of each image is the tongue, and the dark regions are the air spaces of the mouth and throat. The five words are **dean, dune, done, Dan, and Don**, which differ only in their vowel.
    - In **dean**, the tongue is pushed high and forward, almost filling the front of the mouth right up under the roof of the mouth. This leaves a wide dark air space in the throat behind the tongue. The air is squeezed tightly at the front and left open at the back.
    - In **dune**, the tongue is also high but is bunched toward the back of the mouth, and the lips are pushed forward. The narrow squeeze is further back than in "dean," with an open air space in front of the tongue.
    - In **done**, the tongue sits in a middle position, neither very high nor very low, and neither far forward nor far back. The air space above the tongue is moderate in size.
    - In **Dan**, the jaw is lower and the tongue is lower and toward the front, leaving a large air space in the mouth above the tongue.
    - In **Don**, the jaw is open widest and the tongue is low and pulled back toward the throat. The air space in the mouth is large, while the air in the throat behind the tongue is squeezed.
    - Each of these five tongue positions squeezes the air in a different place and by a different amount. By perturbation theory, each shape gives the air a different combination of mass and stiffness in different regions, and therefore a different

set of vibration frequencies. That set of frequencies is what makes "dean" sound different from "Don."

- **The listener has to recover these frequencies from the sound.** For a listener to understand speech, the ear has to figure out which low, middle, and high frequencies are present in the speaker's voice and how strong each one is. The brain then has to use that information to decide which sound was spoken.
    - For example, hearing a high frequency of one particular strength together with a low frequency of another particular strength might tell the brain that the word was **"scene."** Hearing two other frequencies instead might tell the brain that the word was **"son."**
    - The question is how the ear separates a single incoming sound into its separate frequencies and reports each one to the brain. The answer lies in the structure of the inner ear.

Section 4: How We Hear and the Cochlea

- **Sound is a wave of vibration in the air.** When someone speaks, waves of vibration travel through the air and arrive at the listener's **external ear**, the part of the ear on the outside of the head. From there, the vibration travels inward through the ear canal toward the **inner ear**.
- **The cochlea is the part of the inner ear that reports frequencies to the brain.** Of all the structures in the ear, the **cochlea** is the essential one to know by name. It is a small, coiled tube shaped like a snail shell, buried deep inside the bone of the skull. The cochlea is the part that sends signals to the brain saying which frequencies are present in a sound.
- **The inner ear and its parts.** In a cutaway view of the head, the outer ear and ear canal lead inward to a tiny structure boxed off deep in the skull, and that boxed region is the inner ear. Enlarged, the inner ear has two main regions.
    - The upper region contains three looping tubes called the **semicircular ducts**, named anterior, lateral, and posterior, along with two small sacs called the **utricle** and **saccule**. Small sensory structures called **cristae** sit inside swellings at the base of the loops, called **ampullae**. These structures are part of the same inner ear but are not the part that separates sound frequencies.
    - The lower region is the coiled **cochlea**. Running through the cochlea are three fluid-filled channels lying side by side: the **vestibular duct**, the **cochlear duct**, and the **tympanic duct**. These channels spiral together around the coil.
    - The inner ear is built as a set of soft, membrane-walled tubes, called the **membranous labyrinth**, sitting inside a matching set of hollow spaces in hard bone, called the **bony labyrinth**. The soft inner tubes are where the sensing happens, and the bone around them protects them.
    - A bundle of nerve fibers called the **vestibulocochlear nerve** leaves the inner ear and runs toward the brain. This nerve carries the cochlea's signals about sound frequencies to the brain.

Section 5: The Basilar Membrane

- **Sound reaches the cochlea through a chain of structures.** Sound waves in the air first strike the **tympanic membrane**, which is the eardrum. The tympanic membrane is a thin sheet stretched across the end of the ear canal, and it vibrates back and forth as the air pressure pushes on it.
    - The tympanic membrane is attached to the first of three tiny bones, the **malleus**. When the eardrum vibrates, it moves the malleus.
    - The malleus is connected to the second bone, the **incus**, so the malleus's vibration moves the incus.
    - The incus is connected to the third bone, the **stapes**. The stapes sits against an opening in the cochlea, so when the stapes vibrates, it pushes in and out on the fluid inside the cochlea.
    - This chain carries the vibration from the air outside, through the eardrum and three small bones, into the fluid of the inner ear.
- **The basilar membrane runs the length of the cochlea.** Inside the coiled cochlea is a long, thin strip of tissue called the **basilar membrane**. It begins near the stapes, where vibration enters the cochlea, and it follows the spiral all the way to the tip of the coil at the center.
    - The opening at the very tip of the coil is called the **helicotrema**. The basilar membrane extends from the stapes end of the cochlea all the way in to the helicotrema.

    - When the stapes pushes on the cochlear fluid, the fluid moves, and that movement makes the basilar membrane vibrate. The basilar membrane is the structure that turns the vibration of the fluid into information about frequency.

Section 6: How the Basilar Membrane Separates Frequencies

- **The basilar membrane is not the same all along its length.** If we imagine uncoiling the cochlea and laying the basilar membrane out flat as a long strip, one end of the strip is wide and the other end is narrow. The wide end is the part near the helicotrema, and the narrow end is the part near the stapes.
    - At the wide end, the membrane has a **large mass**, labeled M, and a **small stiffness**, labeled k. This end is heavy and floppy.
    - At the narrow end, the membrane has a **small mass**, labeled m, and a **large stiffness**, labeled K. This end is light and stiff.
    - Between the two ends, the mass gradually decreases and the stiffness gradually increases along the length of the membrane.
- **Each spot on the membrane acts like its own mass-spring system.** Every small piece of the basilar membrane has its own mass and its own stiffness, so every piece has its own natural frequency of vibration. Perturbation theory tells us which way that frequency goes along the membrane.
    - At the heavy, floppy end, large mass lowers the frequency and small stiffness also lowers the frequency. Both effects push in the same direction, so this end vibrates at **low frequencies**.
    - At the light, stiff end, small mass raises the frequency and large stiffness also raises the frequency. Both effects again push in the same direction, so this end vibrates at **high frequencies**.
    - Therefore **frequency increases steadily along the membrane**, from low frequencies at the heavy, floppy end to high frequencies at the light, stiff end.
- **A sound with many frequencies is sorted out by place.** When a sound containing several frequencies at once enters the cochlea, each frequency makes the part of the membrane tuned to that frequency vibrate the most. A low-frequency component shakes the heavy, floppy region, and a high-frequency component shakes the light, stiff region.
    - A single complex sound, such as a vowel, is thus spread out along the membrane, with each of its frequencies causing strong vibration at its own location. How strongly each location vibrates reflects how strong that frequency is in the sound.
- **Neurons along the membrane report the pattern to the brain.** Rows of **neurons** are attached along the length of the basilar membrane. When a region of the membrane vibrates, the neurons attached to that region send signals, and those signals travel **to the brain**.
    - The brain can tell which frequencies are present by noticing **which neurons are active**, since each neuron is connected to a location that responds to one particular frequency. It can tell how strong each frequency is by how strongly those neurons are signaling.
    - This is how the listener's brain can register "a high frequency of this much and a low frequency of this much." A pattern of activity at one set of locations corresponds to "scene," and a pattern at a different set of locations corresponds to "son."
- **Georg von Békésy won the Nobel Prize in 1961** for his work on how the cochlea responds to sound. His work established that different frequencies cause the strongest vibration at different places along the basilar membrane, which is the mechanism that lets the ear separate a sound into its frequencies.

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Section 7: Vocal Folds

- **The vocal folds are two small muscles in the throat.** Each vocal fold is about **1 cm in length** and about **1 mm in height**. The folds sit across the airway, and air from the lungs has to pass between them on its way up to the mouth.
    - In a camera view looking down the throat at the vocal folds during speech, the two folds meet along a line, and the gap between them appears as a narrow dark slit. That slit is the opening that air passes through, and it is the space that repeatedly opens and closes when the folds vibrate.

- **Muscle has mass and stiffness, so the vocal folds behave like a mass-spring system.** Since a vocal fold has both mass and stiffness, it can oscillate just like the mass on a spring, and perturbation theory applies to it directly.
    - About **10 other muscles** are attached to the bones and cartilages of the structure that houses the vocal folds. By pulling on that housing, these muscles can change the vocal folds in two ways.
    - They can **stretch** the folds, which makes them **stiffer**. By perturbation theory, greater stiffness raises the frequency of vibration.
    - They can **bunch** the folds together, which makes them **effectively heavier**. By perturbation theory, greater mass lowers the frequency of vibration.
    - In this way, speakers control how fast their vocal folds vibrate by adjusting their effective mass and stiffness.
- **Controlling vocal fold frequency serves several purposes.** Speakers change this frequency constantly, and it carries different kinds of information in different settings.
    - In **singing**, changing the vibration frequency of the vocal folds is how a singer produces different musical notes.
    - In **English**, changes in this frequency convey **questions**, **mistrust**, **degree of belief**, **surprise**, **emotion**, and more. The same words can mean different things depending on how the frequency rises and falls across them.
    - In a **tone language**, changes in this frequency distinguish the **meanings of words**. Two words with identical consonants and vowels can be completely different words if they are spoken with different vocal fold frequencies.
- **Pitch, F0, and fundamental frequency.** The vibration frequency of the vocal folds is called **pitch**. It is also called F_0, which stands for the **fundamental frequency**. A higher F_0 means the vocal folds are opening and closing more times per second, and listeners hear this as a higher pitch.

Section 8: How the Vocal Folds Vibrate

- **Air pressure from the lungs blows the vocal folds apart.** When we take air into our lungs and then close the vocal folds, the air cannot escape, so a lot of pressure builds up beneath the folds. If that pressure becomes high enough, it forces the vocal folds apart, and air begins to flow up through the gap between them.
- **The flowing air sucks the vocal folds back together.** Air rushing through the narrow gap between the folds pulls the folds back inward toward each other. This works just like a curtain in a shower, which gets pulled inward toward the running water.
    - The inward pulling force created by fast-moving air is called the **Bernoulli force**. The same force is what lifts airplanes.
    - Once the Bernoulli force pulls the folds back together, the gap closes and the airflow stops.
- **The cycle repeats hundreds of times per second.** With the folds closed again, pressure builds beneath them once more, and it blows them apart again. The airflow then sucks them back together, and the pressure blows them apart yet again.
    1. Pressure builds beneath the closed vocal folds.
    2. The pressure becomes high enough to blow the folds apart, and air flows between them.
    3. The Bernoulli force from the flowing air sucks the folds back together, closing the gap.
    4. With the gap closed, pressure builds again, and the cycle starts over.
    - This cycle repeats **hundreds of times per second**. Each full cycle of opening and closing is one vibration, so the number of cycles per second is the fundamental frequency F_0. The mass and stiffness of the folds, set by the surrounding muscles, determine how fast this cycle runs.

Section 9: The Harmonic Oscillator Equations

- **A harmonic oscillator is a mass on a spring described with two equations.** The motion of a mass-spring system with no damping is called a **harmonic oscillator**, and it is described by two equations, one for position and one for velocity.

$$\dot{x} = v$$

$$\dot{v} = -\frac{k}{m}\, x$$

- **The dot means rate of change over time.** A dot above a variable means how fast that variable is changing as time passes. So $\dot{x}$ is how fast the position x is changing, and $\dot{v}$ is how fast the velocity v is changing.
    - The first equation, $\dot{x} = v$, says that the rate at which position changes is the velocity. If the mass is moving to the right with some speed, its position increases at exactly that speed.
    - The second equation, $\dot{v} = -\frac{k}{m}x$, says that the rate at which velocity changes depends on the position. The minus sign means the change in velocity always points back toward $x = 0$. When the mass is stretched out to positive x, its velocity is pushed in the negative direction, and when the mass is squeezed to negative x, its velocity is pushed in the positive direction.
    - The ratio $\frac{k}{m}$ sets how strongly the velocity is pushed. A stiffer spring (larger k) pushes harder, and a heavier mass (larger m) is pushed less. This ratio contains both parts of perturbation theory in one quantity.
- **Starting values.** To follow the motion step by step, we pick specific numbers. The ratio is $\frac{k}{m} = 1$, and the time step is $\Delta t = 0.5$. The **time step** Δt is the small chunk of time we move forward with each calculation. The motion starts at $x = 1$ and $v = 0$, meaning the mass is pulled out to $+1$ and is being held still at the moment it is released.

Section 10: Stepping Through Time: Update x First, Then v

- **Each step updates x first and then updates v using the new x.** With $\frac{k}{m} = 1$, the two update rules are the following.

$$x_{\text{new}} = x_{\text{old}} + \Delta t \cdot v_{\text{old}}$$

$$v_{\text{new}} = v_{\text{old}} - \Delta t \cdot x_{\text{new}}$$

- The first rule moves the position forward by the current velocity multiplied by the time step. This is the equation $\dot{x} = v$ applied over a chunk of time Δt.
- The second rule changes the velocity by the position multiplied by the time step, with a minus sign. This is the equation $\dot{v} = -\frac{k}{m}x$ applied over a chunk of time Δt, with $\frac{k}{m} = 1$.
- The key detail is that the velocity update uses the **new** x, the one just calculated in the same step, rather than the old one. This method is called **position-first symplectic Euler**, and the order of the updates matters.
- **Step 1.** Starting from $x = 1$ and $v = 0$, we update x first.

$$x_1 = 1 + 0.5 \times 0 = 1.000$$

- Then we update v using the new x, which is 1.000.

$$v_1 = 0 - 0.5 \times 1.000 = -0.500$$

- The position has not moved yet, since the mass started at rest. However, the spring has begun pushing the velocity in the negative direction.
- **Step 2.** Now x uses the velocity from step 1.

$$x_2 = 1.000 + 0.5 \times (-0.500) = 0.750$$

- Then v uses the new x, which is 0.750.

$$v_2 = -0.500 - 0.5 \times 0.750 = -0.875$$

- The mass has started moving back toward zero, and it is speeding up in the negative direction.
- **Step 3.** The same pattern continues.

$$x_3 = 0.750 + 0.5 \times (-0.875) = 0.3125 \approx 0.313$$

$$v_3 = -0.875 - 0.5 \times 0.3125 = -1.03125 \approx -1.031$$

- All values are displayed rounded to three decimal places, but each later step uses the full-precision numbers. That is why 0.3125 rather than 0.313 is used to calculate v_3.
- **The full lists after 12 steps.** Continuing for 12 steps produces these values, starting with the initial values 1 and 0.
    - Position: $x =$ $[1,\ 1.000,\ 0.750,\ 0.313,\ -0.203,\ -0.668,\ -0.966,\ -1.022,\ -0.823,\ -0.418,\ 0.091,\ 0.578,\ 0.9$

    - Velocity: $v = [0,\ -0.500,\ -0.875,\ -1.031,\ -0.930,\ -0.596,\ -0.113,\ 0.398,\ 0.810,\ 1.019,\ 0.973,\ 0.684,\ 0.2$
- **The results trace out an oscillation.** When the position values are plotted against time, x starts at 1, falls through zero between the third and fourth steps, reaches its lowest value of -1.022 at the seventh step, and climbs back up through zero to 0.920 by the twelfth step. This is one nearly complete swing of the mass, just like the back-and-forth motion of the mass on the spring.
    - The velocity values follow their own wave. v starts at 0, drops to its most negative value of -1.031 at the third step, which is right around when the mass is passing through $x = 0$. It then rises through zero between the sixth and seventh steps, which is right around when the mass reaches its farthest point at -1.022, and it peaks at 1.019 at the ninth step.
    - The mass moves fastest when it passes through the natural position and moves slowest when it reaches its farthest point. This matches the mass-spring behavior where the mass carries its motion through every crossing.
    - The size of the swing stays close to 1 throughout. The position never goes much beyond -1 or $+1$, so the oscillation does not grow or shrink noticeably.
- **This is an approximation.** These values are not exact samples of a perfect oscillation. They are an approximate oscillator produced by stepping forward in chunks of $\Delta t = 0.5$. The method is also different from updating both x and v at the same time from the old values, which gives different numbers.
- **The same calculation in Python.** The 12 steps can be written as a short loop.

```python
x = [1.0]
v = [0.0]
dt = 0.5
for step in range(12):
    x.append(x[-1] + dt*v[-1])
    v.append(v[-1] - dt*x[-1])
```

- The lists `x` and `v` start with the initial values `1.0` and `0.0`. Each time through the loop, `append` adds one new value to the end of a list.
- In Python, `x[-1]` means the last item in the list `x`. In the first line inside the loop, `x[-1]` is the old position and `v[-1]` is the old velocity, so the new position is the old position plus `dt` times the old velocity.
- By the time the second line runs, the new position has already been appended, so `x[-1]` now refers to the **new** x. This is exactly how the velocity update uses the new position.

Section 11: Explicit Euler

- **Explicit Euler updates both x and v from the previous pair.** A second way to step the harmonic oscillator forward is called **explicit Euler**, also called **forward Euler**. Here both new values are calculated from the previous position and the previous velocity, and the newly calculated x is not used when updating v. The settings are $k = 1, m = 1$, and $dt = 0.5$, with starting values $x[0] = 1$ and $v[0] = 0$.

$$x[n] = v[n-1] \cdot dt + x[n-1]$$

$$v[n] = -\frac{k}{m}\, x[n-1] \cdot dt + v[n-1]$$

- The notation $x[n]$ means the position at step n, and $x[n-1]$ means the position at the step just before it. The same goes for $v[n]$ and $v[n-1]$.
- The first equation says the new position equals the previous velocity times the time step, plus the previous position.
- The second equation says the new velocity equals $-\frac{k}{m}$ times the **previous** position times the time step, plus the previous velocity. Both calculations use the previous pair $x[n-1]$ and $v[n-1]$.
- **Steps 1 through 3 worked out.** With $\frac{k}{m} = \frac{1}{1}$, the first three steps are as follows.

$$x[1] = 0 \times 0.5 + 1 = 1$$

$$v[1] = -\tfrac{1}{1} \times 1 \times 0.5 + 0 = -0.500$$

$$x[2] = (-0.500) \times 0.5 + 1 = 0.750$$

$$v[2] = -\tfrac{1}{1} \times 1 \times 0.5 + (-0.500) = -1$$

$$x[3] = (-1) \times 0.5 + 0.750 = 0.250$$

$$v[3] = -\tfrac{1}{1} \times 0.750 \times 0.5 + (-1) = -1.375$$

- The difference from the previous method shows up right away in step 2. Here $v[2]$ uses $x[1] = 1$, the previous position, and comes out to -1. In the position-first method, v_2 used the new position 0.750 and came out to -0.875.
- By step 3, the positions have separated too. Explicit Euler gives $x[3] = 0.250$, while the position-first method gave 0.313.
- **Steps 4 through 12.** The position calculations continue the same way, each using the previous velocity and previous position.

$$\begin{aligned}
x[4] &= (-1.375) \times 0.5 + 0.250 = -0.438 \\
x[5] &= (-1.500) \times 0.5 + (-0.438) = -1.188 \\
x[6] &= (-1.281) \times 0.5 + (-1.188) = -1.828 \\
x[7] &= (-0.688) \times 0.5 + (-1.828) = -2.172 \\
x[8] &= 0.227 \times 0.5 + (-2.172) = -2.059 \\
x[9] &= 1.313 \times 0.5 + (-2.059) = -1.402 \\
x[10] &= 2.342 \times 0.5 + (-1.402) = -0.231 \\
x[11] &= 3.043 \times 0.5 + (-0.231) = 1.290 \\
x[12] &= 3.159 \times 0.5 + 1.290 = 2.869
\end{aligned}$$

- The velocity calculations each use the previous position and previous velocity.

$$\begin{aligned}
v[4] &= -\tfrac{1}{1} \times 0.250 \times 0.5 + (-1.375) = -1.500 \\
v[5] &= -\tfrac{1}{1} \times (-0.438) \times 0.5 + (-1.500) = -1.281 \\
v[6] &= -\tfrac{1}{1} \times (-1.188) \times 0.5 + (-1.281) = -0.688 \\
v[7] &= -\tfrac{1}{1} \times (-1.828) \times 0.5 + (-0.688) = 0.227 \\
v[8] &= -\tfrac{1}{1} \times (-2.172) \times 0.5 + 0.227 = 1.313 \\
v[9] &= -\tfrac{1}{1} \times (-2.059) \times 0.5 + 1.313 = 2.342 \\
v[10] &= -\tfrac{1}{1} \times (-1.402) \times 0.5 + 2.342 = 3.043 \\
v[11] &= -\tfrac{1}{1} \times (-0.231) \times 0.5 + 3.043 = 3.159 \\
v[12] &= -\tfrac{1}{1} \times 1.290 \times 0.5 + 3.159 = 2.514
\end{aligned}$$

- As before, the displayed values are rounded to three decimal places, while every update is carried out with full precision.
- **The full lists after 12 steps.**
  - Position: $x =$ $[1,\ 1.000,\ 0.750,\ 0.250,\ -0.438,\ -1.188,\ -1.828,\ -2.172,\ -2.059,\ -1.402,\ -0.231,\ 1.290,\ 2.$
  - Velocity: $v =$ $[0,\ -0.500,\ -1.000,\ -1.375,\ -1.500,\ -1.281,\ -0.688,\ 0.227,\ 1.313,\ 2.342,\ 3.043,\ 3.159,\ 2.5$
- **With this time step, explicit Euler makes the oscillation grow.** The position still swings down and back up, so the method still produces an oscillation. However, the swings keep getting bigger. The position reaches -2.172 at step 7 and 2.869 at step 12, far beyond the starting value of 1. The velocity grows as well, reaching 3.159 at step 11.
  - A real mass on a spring cannot swing farther than where it started unless something adds energy to it. So this growth is **numerical error**, not a real physical increase in energy. It comes from the method itself when $dt = 0.5$.
  - The position-first method, using the same $\Delta t = 0.5$, kept the swing near a size of 1. The only difference between the two methods is whether the velocity update uses the new position or the previous one, which is why the order of the updates matters so much.
- **Explicit Euler in Python.**

```python
x = [1.0]
v = [0.0]
dt = 0.5
k = m = 1.0
for n in range(1, 13):
```

```
    x.append(v[n-1] * dt + x[n-1])
    v.append(-(k/m) * x[n-1] * dt + v[n-1])
```

- Here the loop counter `n` runs from 1 up to 12. Inside the loop, `x[n-1]` and `v[n-1]` always refer to the previous pair, so even after the new position is appended, the velocity update still uses the old position `x[n-1]`. Each line is a direct copy of one of the two explicit Euler equations.

Section 12: Solving Differential Equations with Arithmetic, Python, and Software 1.0

- **Euler's explicit method solves differential equations using arithmetic.** A **differential equation** is an equation about rates of change, like $\dot{x} = v$ and $\dot{v} = -\frac{k}{m}x$. Euler's explicit method turns each of these into an update of the form $x[n] = \ldots dt + x[n-1]$, where the new value equals the previous value plus a rate multiplied by a small time step.
  - Every update is just multiplication and addition. By repeating these simple arithmetic steps many times, we can follow how a system like a mass on a spring moves through time, without any other kind of mathematics.
  - The method is named for **Leonhard Euler**, the mathematician whose portrait appears alongside photographs of a woman and of her receiving a medal at a formal ceremony, honoring work carried out with arithmetic methods of this kind.
- **The same stepping procedure is done in Python.** Computers are built to repeat arithmetic quickly and exactly, so a loop in Python can carry out the Euler updates for as many steps as we want. Each pass through the loop computes one new position and one new velocity and appends them to lists, just as in the two loops written for the harmonic oscillator.
- **Software 1.0 is software written as explicit instructions. Software 1.0** refers to programs where people write out every instruction the computer must follow, step by step. The Euler loops are an example, since every arithmetic operation the computer performs is spelled out by the programmer.
  - A black-and-white photograph shows a woman standing beside a stack of printed program code about as tall as she is. Each page in that stack holds lines of instructions written by hand for a computer to follow, which gives a sense of how much written code large Software 1.0 programs can require.
  - A second photograph shows the same woman receiving a medal at a formal ceremony, at a podium bearing a presidential seal, recognizing the importance of that kind of hand-written software.

Course: LING285: Human Language and Technology
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Professor: Khalil Iskarous

Section 1: A Spring Alone Versus a Mass-Spring System

- **A spring stops at a place, but a mass stops after a process.** A spring with no mass attached returns to one location and stays there. A spring with a mass attached keeps moving past that location, back and forth, before it finally comes to rest.
  - Motion is measured with the **displacement** x, which is how far the end of the spring is from its **natural position**. The natural position is where the spring rests when nothing pulls or pushes on it, and it is called $x = 0$. A stretch is $x = +1$, and a squeeze is $x = -1$.
- **A spring alone never crosses its natural position.** A spring fixed to a wall with nothing on its free end, when pulled to $x = +1$ and released, pulls itself back and stops at $x = 0$. Over time, x drops quickly at first and then more and more slowly, getting closer to 0 without ever crossing it.
  - The motion slows down as the spring nears $x = 0$, since the spring's pull depends on how far it is stretched. A large stretch produces a large pull back, and a small stretch produces a small pull, so near $x = 0$ the pull is almost zero.
  - When the spring is squeezed to $x = -1$ and released, the same thing happens in the opposite direction. x rises toward 0, quickly at first and then slowly, and never goes above it.
  - Whether it starts at $+1$ or -1, a massless spring always ends at the same place without passing through it. This is what it means for a spring to **stop at a place**.
- **A mass on the spring keeps moving through the natural position.** When a mass is attached to the spring, pulled to $x = +1$, and released, it does not stop at $x = 0$. It passes

right through the natural position and keeps going to the other side.
    - The mass carries its motion with it. When it reaches $x = 0$, the spring is no longer pulling on it, but the mass is already moving fast, so it continues past 0 and compresses the spring.
    - The compressed spring pushes the mass back. The mass slows, stops briefly near $x = -1$, and is pushed back through $x = 0$ toward the positive side. The mass **carries its motion through every crossing**.
    - Over time, x forms a wave. It starts at $+1$, falls through 0 to nearly -1, rises back through 0 to nearly $+1$, and repeats. This repeating back-and-forth motion is called **oscillation**.
- **Lightly damped oscillation.** Each swing of the mass is slightly smaller than the one before, so the peaks of the wave shrink a little over time. This is a **lightly damped oscillation**, meaning the motion is slowly drained away but the mass still swings many times before settling.
    - The mass therefore **stops after a process**. It eventually rests at $x = 0$, but only after a long series of crossings, each a little smaller than the last.
    - A mass-spring system has a **frequency**, the number of back-and-forth cycles it completes per second. A spring alone has no frequency, since it never swings.

Section 2: Perturbation Theory

- **Perturbation theory** asks which direction a system's behavior changes when one of its properties is changed. For a mass-spring system, the two properties are the **mass** m and the **stiffness** k, and the behavior is the **frequency** f of the oscillation.
- **Increasing the mass decreases the frequency.** A heavier mass on the same spring oscillates more slowly.
    - A heavier mass is harder to speed up and harder to slow down. The spring's pull changes its speed less, so each swing takes longer and fewer swings fit into one second. When m goes up, f goes down.
- **Increasing the stiffness increases the frequency.** The same mass on a stiffer spring oscillates faster.
    - **Stiffness** k is how strongly a spring resists being stretched or squeezed. A stiffer spring pulls back harder for the same stretch, so it speeds the mass up faster, each swing finishes sooner, and more swings fit into one second. When k goes up, f goes up.
- Together, these rules are the core of perturbation theory for oscillators: **more mass means lower frequency, and more stiffness means higher frequency**. No exact frequency needs to be calculated to use these rules, only the direction of the change.

Section 3: Speech as Changing Mass and Stiffness of Air

- **Speaking changes the effective mass and stiffness of air in the vocal tract.** The **vocal tract** is the air-filled tube from the vocal folds in the throat, through the mouth, to the lips. Moving the tongue, jaw, and lips squeezes some portions of that air and opens up others.
    - Squeezing a portion of air changes how much it acts like a mass and how much it acts like a spring. The **effective mass** and **effective stiffness** of each portion of air are different for every vowel and every consonant, since each sound gives the tube a different shape.
    - Vocal fold vibration sets the air in the vocal tract vibrating. Since the air has different mass and stiffness in different places, it vibrates at **many frequencies at once**. This set of frequencies depends on **where** the air is squeezed and **how much**, so each vowel and consonant has its own set.
- **Each vowel squeezes the air in a different place.** In side-view MRI slices through the middle of the head, the tongue appears as a bright rounded mass and the air spaces of the mouth and throat appear dark. The words **dean, dune, done, Dan, and Don** differ only in their vowel, and each vowel has a different tongue position.
    - In **dean**, the tongue is high and forward, almost filling the front of the mouth under its roof, with a wide air space in the throat behind the tongue. The air is squeezed at the front and open at the back.
    - In **dune**, the tongue is high but bunched toward the back of the mouth, and the lips are pushed forward. The squeeze is further back than in "dean," with open air space in front of the tongue.
    - In **done**, the tongue sits in a middle position, neither very high nor very low and neither far forward nor far back.

    - In **Dan**, the jaw and tongue are lower, with the tongue toward the front, leaving a large air space above the tongue.
    - In **Don**, the jaw is open widest and the tongue is low and pulled back, leaving a large air space in the mouth while squeezing the air in the throat behind the tongue.
    - By perturbation theory, each tongue position gives the air a different combination of mass and stiffness in different regions, and therefore a different set of vibration frequencies. That set of frequencies distinguishes one vowel from another.
- **The listener must recover these frequencies.** To understand speech, the ear has to determine which low, middle, and high frequencies are in the speaker's voice and how strong each is. The brain then uses that information to decide what was said.
    - For example, a high frequency of one strength combined with a low frequency of another strength may signal the word **"scene,"** while two other frequencies signal the word **"son."** The ear must separate one incoming sound into its individual frequencies and report each to the brain.

Section 4: How We Hear and the Cochlea

- **Sound is a wave of vibration in the air.** Waves of vibration from a speaker travel through the air, arrive at the listener's **external ear**, and travel inward to the **inner ear**.
- **The cochlea sends frequency information to the brain.** The **cochlea** is the part of the inner ear that sends signals to the brain saying which frequencies are present in a sound. It is the one anatomical name in the ear that is essential to know. The cochlea is a small coiled tube, shaped like a snail shell, deep within the skull.
- **The inner ear has two main regions.** The inner ear sits deep in the skull, inward from the external ear and ear canal.
    - The upper region contains three looping **semicircular ducts**, named anterior, lateral, and posterior, along with two sacs called the **utricle** and **saccule**. Structures called **cristae** sit within swellings at the bases of the loops, called **ampullae**.
    - The lower region is the coiled **cochlea**. Three channels run side by side through it and spiral together around the coil: the **vestibular duct**, the **cochlear duct**, and the **tympanic duct**.
    - The inner ear consists of soft membrane-walled tubes, the **membranous labyrinth**, sitting inside matching hollow spaces in bone, the **bony labyrinth**.
    - The **vestibulocochlear nerve** leaves the inner ear and runs toward the brain, carrying the cochlea's signals.

Section 5: The Basilar Membrane

- **Vibration reaches the cochlea through a chain of structures.** Sound waves first strike the **tympanic membrane**, the eardrum, which vibrates as the air pressure pushes on it.
    - The tympanic membrane is attached to the first of three tiny bones, the **malleus**, so its vibration moves the malleus.
    - The malleus is connected to the second bone, the **incus**, and moves it.
    - The incus is connected to the third bone, the **stapes**. The stapes sits against the cochlea, so its vibration is passed into the cochlea.
- **The basilar membrane runs along the cochlea.** Inside the coiled cochlea is a long, thin strip of tissue called the **basilar membrane**. It begins near the stapes, where vibration enters, and follows the spiral to the center of the coil, where the **helicotrema** is located.
    - Vibration passed in by the stapes makes the basilar membrane vibrate. The basilar membrane is the structure that turns this vibration into information about frequency.

Section 6: How the Basilar Membrane Separates Frequencies

- **The basilar membrane changes in mass and stiffness along its length.** When the basilar membrane is uncoiled and laid out flat as a strip, one end is wide and the other end is narrow.
    - The wide end has a **large mass** M and a **small stiffness** k, so it is heavy and floppy.
    - The narrow end has a **small mass** m and a **large stiffness** K, so it is light and stiff.

    - Between the two ends, mass gradually decreases and stiffness gradually increases.
- **Each spot on the membrane acts as its own mass-spring system.** Each small piece of the basilar membrane has its own mass and stiffness, so each piece has its own frequency of vibration. Perturbation theory sets the direction of that frequency along the membrane.
    - At the heavy, floppy end, large mass lowers the frequency and small stiffness also lowers it. Both effects work together, so this end vibrates at **low frequencies**.
    - At the light, stiff end, small mass raises the frequency and large stiffness also raises it. Both effects work together, so this end vibrates at **high frequencies**.
    - **Frequency increases steadily along the membrane**, from the heavy, floppy end to the light, stiff end.
- **A sound with many frequencies is sorted by place.** When a sound containing several frequencies enters the cochlea, each frequency makes the part of the membrane matched to that frequency vibrate most. Low-frequency parts of the sound shake the heavy, floppy region, and high-frequency parts shake the light, stiff region.
    - A single complex sound, such as a vowel, is spread out along the membrane, with each of its frequencies producing strong vibration at its own location. How strongly each location vibrates reflects how strong that frequency is in the sound.
- **Neurons along the membrane carry the pattern to the brain. Neurons** are attached along the length of the basilar membrane. When a region vibrates, the neurons attached to it send signals **to the brain**.
    - The brain identifies which frequencies are present by which neurons are active, since each neuron is attached to a location that responds to a particular frequency. It identifies how strong each frequency is by how strongly those neurons signal.
    - In this way the brain registers "a high frequency of this much and a low frequency of this much." One pattern of activity along the membrane corresponds to "scene," and a different pattern corresponds to "son."
- **Georg von Békésy** won the **Nobel Prize in 1961** for explaining how hearing works in this way.

Section 7: Vocal Folds

- **The vocal folds are small muscles.** Each vocal fold is about **1 cm in length** and **1 mm in height**. The two folds sit across the airway, and during speech the gap between them is a narrow slit that air passes through.
- **Muscle has mass and stiffness, so the vocal folds oscillate like a mass-spring system.** Perturbation theory applies to them directly.
    - About **10 other muscles** are attached to the bones and cartilages of the housing of the vocal folds. These muscles change the folds in two ways.
    - They can **stretch** the folds, making them **stiffer**, which raises the frequency of vibration.
    - They can **bunch** the folds, making them **effectively heavier**, which lowers the frequency of vibration.
- **Changing vocal fold frequency serves several purposes.**
    - In **singing**, it produces different notes.
    - In **English**, it conveys **questions**, **mistrust**, **degree of belief**, **surprise**, **emotion**, and more.
    - In a **tone language**, it distinguishes the **meanings of words**.
- **Pitch, F_0, and fundamental frequency.** The vibration frequency of the vocal folds is called **pitch**, also written F_0, the **fundamental frequency**. A higher F_0 means the folds open and close more times per second.

Section 8: How the Vocal Folds Vibrate

- **Air pressure blows the vocal folds apart.** When we take air into the lungs and close the vocal folds, a lot of pressure builds beneath them. If the pressure is high enough, it blows the folds apart, and air flows between them.
- **The airflow sucks the vocal folds back together.** Just like a curtain in a shower, the flowing air sucks the vocal folds back inward. This suction force is called the **Bernoulli force**, the same force that lifts airplanes.

- Once the folds are pulled together, the airflow stops, and pressure builds beneath them again. The pressure blows them apart again, the airflow sucks them back in again, and the cycle repeats **hundreds of times per second**.
- Each opening and closing is one vibration, so the number of cycles per second is F_0. The mass and stiffness of the folds, set by the surrounding muscles, determine how fast this cycle runs.

Section 9: The Harmonic Oscillator Equations

- **The harmonic oscillator is a mass on a spring described by two equations,** one for position x and one for velocity v.

$$\dot{x} = v$$

$$\dot{v} = -\frac{k}{m}\,x$$

- **A dot above a variable means its rate of change over time.** $\dot{x}$ is how fast position changes, and $\dot{v}$ is how fast velocity changes.
  - The first equation says that position changes at a rate equal to the velocity.
  - The second equation says that velocity changes at a rate set by the position. The minus sign means this change always points back toward $x = 0$. At positive x, velocity is pushed negative, and at negative x, velocity is pushed positive.
  - The ratio $\frac{k}{m}$ sets how strongly velocity is pushed. Larger stiffness k pushes harder, and larger mass m is pushed less, so both rules of perturbation theory are contained in this one ratio.
- **Settings for a step-by-step calculation.** The ratio is $\frac{k}{m} = 1$, and the **time step** is $\Delta t = 0.5$, the chunk of time moved forward with each calculation. The motion starts at $x = 1$ and $v = 0$, so the mass is pulled to $+1$ and held still at release.

Section 10: Stepping Through Time: Update x First, Then v

- **Each step updates x first, then updates v using the new x.** With $\frac{k}{m} = 1$, the update rules are the following.

$$x_{\text{new}} = x_{\text{old}} + \Delta t \cdot v_{\text{old}}$$

$$v_{\text{new}} = v_{\text{old}} - \Delta t \cdot x_{\text{new}}$$

- The first rule moves the position forward by the velocity times the time step, which is $\dot{x} = v$ applied over Δt.
- The second rule changes the velocity by minus the position times the time step, which is $\dot{v} = -\frac{k}{m}x$ applied over Δt.
- The velocity update uses the **new** x just calculated in the same step. This method is called **position-first symplectic Euler**, and the **order of the updates matters**.
- **Step 1.** Starting from $x = 1$ and $v = 0$, position is updated first.

$$x_1 = 1 + 0.5 \times 0 = 1.000$$

- Velocity is then updated with the new position 1.000.

$$v_1 = 0 - 0.5 \times 1.000 = -0.500$$

- The position has not moved yet, since the mass started at rest, but the velocity has begun to turn negative.
- **Step 2.**

$$x_2 = 1.000 + 0.5 \times (-0.500) = 0.750$$

$$v_2 = -0.500 - 0.5 \times 0.750 = -0.875$$

- The mass is moving back toward zero and speeding up in the negative direction.
- **Step 3.**

$$x_3 = 0.750 + 0.5 \times (-0.875) = 0.3125 \approx 0.313$$

$$v_3 = -0.875 - 0.5 \times 0.3125 = -1.03125 \approx -1.031$$

- Values are displayed to three decimal places, but later steps use full precision, so v_3 is computed with 0.3125 rather than 0.313.
- **Values after 12 steps**, including the starting values:
  - Position: $x =$ $[1,\ 1.000,\ 0.750,\ 0.313,\ -0.203,\ -0.668,\ -0.966,\ -1.022,\ -0.823,\ -0.418,\ 0.091,\ 0.578,\ 0.9$
  - Velocity: $v =$ $[0,\ -0.500,\ -0.875,\ -1.031,\ -0.930,\ -0.596,\ -0.113,\ 0.398,\ 0.810,\ 1.019,\ 0.973,\ 0.684,\ 0.2$
- **These values form an oscillation.** Position starts at 1, crosses zero between steps 3 and 4, reaches its lowest value of -1.022 at step 7, and climbs back through zero to 0.920 by step 12, which is nearly one complete swing.
  - Velocity reaches its most negative value of -1.031 at step 3, around when the mass passes through $x = 0$. It crosses zero between steps 6 and 7, around when the mass reaches its farthest point, and peaks at 1.019 at step 9.
  - The mass moves fastest as it passes through the natural position and slowest at its farthest points, matching the mass-spring motion from Section 1.
  - The size of the swing stays close to 1 throughout, so the oscillation neither grows nor shrinks noticeably.
- **This is an approximate oscillator.** The values are not exact points of a perfect oscillation, since they come from stepping forward in chunks of $\Delta t = 0.5$. The method also gives different numbers from updating x and v at the same time from the old values.
- **The same calculation in Python:**

```python
x = [1.0]
v = [0.0]
dt = 0.5
for step in range(12):
    x.append(x[-1] + dt*v[-1])
    v.append(v[-1] - dt*x[-1])
```

- The lists `x` and `v` start with `1.0` and `0.0`, and `append` adds one new value to the end of a list on each pass through the loop.
- `x[-1]` is the last item in the list `x`. In the first line of the loop, `x[-1]` and `v[-1]` are the old position and velocity.
- By the second line, the new position has already been appended, so `x[-1]` is now the **new** x, which is how the velocity update uses the new position.

Section 11: Explicit Euler

- **Explicit Euler updates both x and v from the previous pair.** In **explicit Euler**, also called **forward Euler**, both new values are computed from the previous position and previous velocity, and the new x is not used to update v. The settings are $k = 1, m = 1$, and $dt = 0.5$, with $x[0] = 1$ and $v[0] = 0$.

$$x[n] = v[n-1] \cdot dt + x[n-1]$$

$$v[n] = -\frac{k}{m}\, x[n-1] \cdot dt + v[n-1]$$

- $x[n]$ is the position at step n, and $x[n-1]$ is the position at the step before; the same holds for v.
- The new position is the previous velocity times dt, plus the previous position. The new velocity is $-\frac{k}{m}$ times the **previous** position times dt, plus the previous velocity. **Both calculations use the previous pair** $x[n-1]$ and $v[n-1]$.
- **Steps 1 through 3**, with $\frac{k}{m} = \frac{1}{1}$:

$$x[1] = 0 \times 0.5 + 1 = 1$$

$$v[1] = -\tfrac{1}{1} \times 1 \times 0.5 + 0 = -0.500$$

$$x[2] = (-0.500) \times 0.5 + 1 = 0.750$$

$$v[2] = -\tfrac{1}{1} \times 1 \times 0.5 + (-0.500) = -1$$

$$x[3] = (-1) \times 0.5 + 0.750 = 0.250$$

$$v[3] = -\tfrac{1}{1} \times 0.750 \times 0.5 + (-1) = -1.375$$

- The two methods separate at step 2. Explicit Euler computes $v[2]$ from the previous position $x[1] = 1$ and gets -1, while the position-first method used the new position 0.750 and got -0.875.
- By step 3 the positions differ too: explicit Euler gives 0.250, while the position-first method gave 0.313.
- **Steps 4 through 12, positions:**

$$\begin{aligned}
x[4] &= (-1.375) \times 0.5 + 0.250 = -0.438 \\
x[5] &= (-1.500) \times 0.5 + (-0.438) = -1.188 \\
x[6] &= (-1.281) \times 0.5 + (-1.188) = -1.828 \\
x[7] &= (-0.688) \times 0.5 + (-1.828) = -2.172 \\
x[8] &= 0.227 \times 0.5 + (-2.172) = -2.059 \\
x[9] &= 1.313 \times 0.5 + (-2.059) = -1.402 \\
x[10] &= 2.342 \times 0.5 + (-1.402) = -0.231 \\
x[11] &= 3.043 \times 0.5 + (-0.231) = 1.290 \\
x[12] &= 3.159 \times 0.5 + 1.290 = 2.869
\end{aligned}$$

- **Steps 4 through 12, velocities:**

$$\begin{aligned}
v[4] &= -\tfrac{1}{1} \times 0.250 \times 0.5 + (-1.375) = -1.500 \\
v[5] &= -\tfrac{1}{1} \times (-0.438) \times 0.5 + (-1.500) = -1.281 \\
v[6] &= -\tfrac{1}{1} \times (-1.188) \times 0.5 + (-1.281) = -0.688 \\
v[7] &= -\tfrac{1}{1} \times (-1.828) \times 0.5 + (-0.688) = 0.227 \\
v[8] &= -\tfrac{1}{1} \times (-2.172) \times 0.5 + 0.227 = 1.313 \\
v[9] &= -\tfrac{1}{1} \times (-2.059) \times 0.5 + 1.313 = 2.342 \\
v[10] &= -\tfrac{1}{1} \times (-1.402) \times 0.5 + 2.342 = 3.043 \\
v[11] &= -\tfrac{1}{1} \times (-0.231) \times 0.5 + 3.043 = 3.159 \\
v[12] &= -\tfrac{1}{1} \times 1.290 \times 0.5 + 3.159 = 2.514
\end{aligned}$$

- Displayed values are rounded to three decimal places, and every update uses full precision.
- **Values after 12 steps:**
  - Position: $x =$ $[1,\ 1.000,\ 0.750,\ 0.250,\ -0.438,\ -1.188,\ -1.828,\ -2.172,\ -2.059,\ -1.402,\ -0.231,\ 1.290,\ 2.$
  - Velocity: $v =$ $[0,\ -0.500,\ -1.000,\ -1.375,\ -1.500,\ -1.281,\ -0.688,\ 0.227,\ 1.313,\ 2.342,\ 3.043,\ 3.159,\ 2.5$
- **With $dt = 0.5$, explicit Euler makes the oscillation grow.** Position still swings down and back up, but the swings get larger, reaching -2.172 at step 7 and 2.869 at step 12, far beyond the starting value of 1. Velocity also grows, reaching 3.159 at step 11.
  - This growth is **numerical error**, not a real physical increase in energy. It comes from the method itself at this time step.
  - The position-first method, with the same $\Delta t = 0.5$, kept the swing near a size of 1. The only difference between the methods is whether the velocity update uses the new position or the previous one.
- **Explicit Euler in Python:**

```python
x = [1.0]
v = [0.0]
dt = 0.5
k = m = 1.0
for n in range(1, 13):
    x.append(v[n-1] * dt + x[n-1])
    v.append(-(k/m) * x[n-1] * dt + v[n-1])
```

- The counter `n` runs from 1 to 12. `x[n-1]` and `v[n-1]` always refer to the previous pair, so the velocity update uses the old position even after the new position has been appended. Each line inside the loop is one of the two explicit Euler equations.

Section 12: Solving Differential Equations with Arithmetic, Python, and Software 1.0

- **Euler's explicit method solves differential equations using arithmetic.** A **differential equation** is an equation about rates of change, such as $\dot{x} = v$ and $\dot{v} = -\frac{k}{m}x$. The method, named for **Leonhard Euler**, turns each one into an update of the form $x[n] = \ldots dt + x[n-1]$, in which the new value is the previous value plus a rate times a small time step.
    - Each update uses only multiplication and addition. Repeating these arithmetic steps follows the motion of a system, such as a mass on a spring, through time. Work done with arithmetic methods of this kind has been honored with a medal at a formal ceremony.
- **The Euler steps are carried out in Python.** A Python loop repeats the arithmetic for as many steps as needed, computing one new position and one new velocity on each pass and appending them to lists.
- **Software 1.0** is software in which people write out every instruction the computer follows, step by step. The Euler loops are an example, since every arithmetic operation is spelled out by the programmer.
    - Large Software 1.0 programs filled stacks of printed code as tall as the person who wrote them, and that hand-written software was honored with a medal presented at a presidential ceremony.